University of the Philippines Diliman
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Physics 180 THV-1
Problem Set 4-5

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IV - BS Physics

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Problem 4.1

There are situations when a radioactive decay results in a product nucleus that is also radioactive. We can have a decay chain 1 to 2 to 3 and so on, with the original nucleus (Type 1) being the parent. Assume that we begin with N_0 atoms of the parent (i.e., $N_1(0) = N_0$) and no decay products are originally present (i.e., $N_i(0) = 0, i = 2, 3, \ldots$). Refer to the decay constants as $\lambda_1, \lambda_2, \lambda_3, \ldots$

We now consider the decay 1 to 2 to 3, where nuclei of type 3 are stable.

(a) Set up and solve the differential equation for $N_3(t)$.

(b) Evaluate $N_1(t) + N_2(t) + N_3(t)$ and interpret.

(c) Examine $N_1(t), N_2(t), N_3(t)$ at small t and interpret.

(d) Find the limits of $N_1(t), N_2(t), N_3(t)$ as t approaches infinity and interpret.

Solution

First, we set up the complete coupled differential equations for each product in the decay chain, starting with the parent type 1 nucleus, which follows a typical decay rate

$$\frac{dN_1}{dt} = -\lambda_1 N_1 \tag{1.1}$$

For the type 2 nucleus, it is being produced by the decay of the type 1 nucleus, so it has a $+\lambda_1 N_1$ term. However, it also decays at a rate $\lambda_2 N_2$, hence:

$$\frac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2 \tag{1.2}$$

As for the type 3 nucleus, which is stable, it is being produced completely only by type two nucleus decay, and as such

$$\frac{dN_3}{dt} = \lambda_2 N_2 \tag{1.3}$$

Those are the coupled rate equations. Now, we begin solving for the given problems.

(a) Set up and solve the differential equation for $N_3(t)$.

First we solve for the parent equation (1.1), it has solution

$$N_1(t) = N_0 e^{-\lambda_1 t} \tag{1.4}$$

Substuting this into (1.2), we have

$$\frac{dN_2}{dt} = \lambda_1 N_0 e^{-\lambda_1 t}$$

Rearranging, we find

$$\frac{dN_2}{dt} + \lambda_2 N_2 = \lambda_1 N_0 e^{-\lambda_1 t} \tag{1.5}$$

This is a first order linear ODE, and can be solved with an integrating factor. By inspection of the coefficient of the second term, we find that the integrating factor is $e^{\lambda_2 t}$, and as such

$$\frac{d}{dt}\left(N_2 e^{\lambda_2 t}\right) = \lambda_1 N_0 e^{(\lambda_2-\lambda_1)t} \tag{1.6}$$

Multiplying by dt and integrating, we find

$$\begin{aligned}
d(N_2 e^{\lambda_2 t}) &= \lambda_1 N_0 e^{(\lambda_2-\lambda_1)t}\,dt \\
\int_0^t d(N_2 e^{\lambda_2 t}) &= \int_0^t \lambda_1 N_0 e^{(\lambda_2-\lambda_1)t}\,dt \\
N_2 e^{\lambda_2 t} - N_2(0) &= \frac{\lambda_1 N_0}{\lambda_2-\lambda_1}\left(e^{(\lambda_2-\lambda_1)t} - e^{(\lambda_2-\lambda_1)0}\right) \\
N_2 e^{\lambda_2 t} &= \frac{\lambda_1 N_0}{\lambda_2-\lambda_1}\left(e^{(\lambda_2-\lambda_1)t} - 1\right)
\end{aligned}$$

Hence, the solution for $N_2(t)$ is

$$N_2(t) = \frac{\lambda_1 N_0}{\lambda_2-\lambda_1}\left(e^{-\lambda_1 t} - e^{-\lambda_2 t}\right) \tag{1.7}$$

Now, we use (1.3)

$$\frac{dN_3}{dt} = \lambda_2 N_2$$

This has general solution[1]

$$N_3(t) = \lambda_2 \int_0^t N_2(t)\,dt$$

Substituting (1.7), we have

$$N_3(t) = \frac{\lambda_1\lambda_2 N_0}{\lambda_2-\lambda_1}\int_0^t (e^{-\lambda_1 t} - e^{-\lambda_2 t})\,dt$$

Since

$$\int_0^t e^{-\lambda t}\,dt = \frac{1-e^{-\lambda t}}{\lambda},$$

Then

$$N_3(t) = \frac{\lambda_1\lambda_2 N_0}{\lambda_2-\lambda_1}\left(\frac{1-e^{-\lambda_1 t}}{\lambda_1} - \frac{1-e^{-\lambda_2 t}}{\lambda_2}\right)$$

Simplifying, we have

$$\boxed{N_3(t) = \frac{\lambda_1\lambda_2 N_0}{\lambda_2-\lambda_1}\left(1 - \frac{\lambda_2 e^{-\lambda_1 t} - \lambda_1 e^{-\lambda_2 t}}{\lambda_2-\lambda_1}\right)} \tag{1.8}$$

[1] I don't change variables of integration, it's annoying

(b) Evaluate $N_1(t) + N_2(t) + N_3(t)$ and interpret. Adding together (1.4), (1.7), and (1.8), we find

$$\begin{aligned}
N_1(t) + N_2(t) + N_3(t) &= N_0 e^{-\lambda_1 t} + \frac{\lambda_1 N_0}{\lambda_2 - \lambda_1}\left(e^{(-\lambda_1)t} - e^{-\lambda_2 t}\right) + \frac{\lambda_1\lambda_2 N_0}{\lambda_2 - \lambda_1}\left(\frac{1 - e^{-\lambda_1 t}}{\lambda_1} - \frac{1 - e^{-\lambda_2 t}}{\lambda_2}\right) \\
&= N_0\left(e^{-\lambda_1 t} + \frac{\lambda_1}{\lambda_2 - \lambda_1}\left(e^{-\lambda_1 t} - e^{-\lambda_2 t}\right) + \frac{\lambda_1\lambda_2}{\lambda_2 - \lambda_1}\left(\frac{1 - e^{-\lambda_1 t}}{\lambda_1} - \frac{1 - e^{-\lambda_2 t}}{\lambda_2}\right)\right) \\
&= N_0 \frac{1}{\lambda_2 - \lambda_1}\left((\lambda_2 - \lambda_1)e^{-\lambda_1 t} + \lambda_1\left(e^{-\lambda_1 t} - e^{-\lambda_2 t}\right) + \lambda_1\lambda_2\left(\frac{1 - e^{-\lambda_1 t}}{\lambda_1} - \frac{1 - e^{-\lambda_2 t}}{\lambda_2}\right)\right) \\
&= N_0 \frac{1}{\lambda_2 - \lambda_1}\left(\lambda_2 e^{-\lambda_1 t} - \lambda_1 e^{-\lambda_1 t} + \lambda_1 e^{-\lambda_1 t} - \lambda_1 e^{-\lambda_2 t} + \lambda_2 - \lambda_2 e^{-\lambda_1 t} - \lambda_1 + \lambda_1 e^{-\lambda_2 t}\right) \\
&= N_0 \frac{1}{\lambda_2 - \lambda_1}(\lambda_2 - \lambda_1) \\
&= N_0
\end{aligned}$$

Hence

$$\boxed{N_1(t) + N_2(t) + N_3(t) = N_0} \tag{1.9}$$

We can interpret this as the number of nuclei present on the sample is constant over all of time, i.e., $\boxed{\textbf{the number of nuclei remains constant.}}$

(c) Examine $N_1(t), N_2(t)$, and $N_3(t)$ at small t and interpret.

We can taylor expand $e^{-\lambda}t$ as

$$e^{-\lambda t} = 1 - \lambda t + \frac{1}{2}\lambda^2 t^2 \cdots$$

Thus, to first order, and $t \ll 1$, (1.4) reduces to

$$\boxed{N_1 \approx N_0(1 - \lambda_1 t)}$$

That is, the type 1 nuclei decreases linearly.

For (1.7), we find that, to first order

$$e^{-\lambda_1 t} - e^{-\lambda_2 t} \approx (\lambda_2 - \lambda_1)t$$

Substituting this in (1.7), we find that

$$\boxed{N_2 \approx \lambda_1 N_0 t}$$

So, initially, the type 2 nuclei grows linearly.

Using the same argument for the type 3 nuclei, we find that[2]

$$\lambda_2 e^{-\lambda_1 t} - \lambda_1 e^{-\lambda_2 t} = (\lambda_2 - \lambda_1) - \frac{1}{2}\lambda_1\lambda_2(\lambda_2 - \lambda_1)t^2$$

Hence, substituting this in (1.8), we find that

$$\boxed{N_3 \approx \frac{1}{2}\lambda_1\lambda_2 N_0 t^2}$$

[2] I find the entire derivation for this one long and cumbersome but mathematica said that's the reduction, so yeah, just trust me that (1.8) reduces this way.

So, type 3 nuclei grow quadratically.

Hence, we find the following interpretations: At $t = 0$, only type 1 exists. At short time t after $t = 0$, we find that type 1 decreases linearly, and type 2 grows linearly due to that decrease. However, type 3 nuclei need two decay events to form, so it's population initially grows as t^2 rather than linearly.

(d) Find the limits of $N_1(t), N_2(t), N_3(t)$ as t approaches infinity and interpret.

At $t \to \infty$, the following exponential terms tend to go to 0:

$$e^{-\lambda_1 t} \to 0$$

and

$$e^{-\lambda_2 t} \to 0$$

Therefore, evaluating (1.4), (1.7), and (1.8), we find that for type 1, the decay goes to:

$$\boxed{\lim_{t\to\infty} N_1(t) = 0}$$

For type 2, we have

$$\boxed{\lim_{t\to\infty} N_2(t) = 0}$$

And for type 3,

$$\boxed{\lim_{t\to\infty} N_3(t) = N_0}$$

We interpret this as: Over long time scales, all parent and secondary nuclei eventually decays to stable daughter nuclei. The intermediate nucleus does not accumulate indefinitely. It reaches some peak population and eventually disappears as it decays to the stable product. Eventually, all the present nuclei become the stable daughter nuclei. See Fig. 1 in the appendices to see how the population of the decay chain evolves over time.

Problem 5.1

The energy of an alpha particle in terms of the Q-value was obtained by assuming a nonrelativistic form for the kinetic energy. Using ***relativistic expressions*** for p and T, **derive the relativistic version** of the equation we have seen in the lecture

$$T_\alpha = \frac{M_D}{M_\alpha + M_D} Q = \frac{1}{1 + \frac{M_\alpha}{M_D}} Q \tag{2.1}$$

and calculate the error made by neglecting these corrections.

Solution

This solution is pretty long, so it can be broken up into several parts.

5.1.1 Energy Conservation

Assuming a zero momentum nuclei, we find that the momenta of the product nuclei is equal to zero:

$$p_\alpha + p_D = 0 \implies p_\alpha = p_D = p$$

This remains true relativistically.

For a particle of rest mass M and momentum p, we find that

$$E^2 = p^2c^2 + M^2c^4 \tag{2.2}$$

and the kinetic energy is the total energy minus the rest energy,

$$T = E - Mc^2$$

Therefore, the kinetic energy can be given as

$$T = \sqrt{p^2c^2 + M^2c^4} - Mc^2$$

So for the two produces, the daugther nuclei and the alpha particle, we find that

$$T_\alpha = \sqrt{p^2c^2 + M_\alpha^2c^4} - M_\alpha c^2 \tag{2.3}$$

$$T_D = \sqrt{p^2c^2 + M_D^2c^4} - M_D c^2 \tag{2.4}$$

Hence, the exact kinetic energy conservation equation is

$$Q = \sqrt{p^2c^2 + M_\alpha^2c^4} + \sqrt{p^2c^2 + M_D^2c^4} - (M_\alpha + M_D)c^2 \tag{2.5}$$

We introduce the total energy available to the products

$$E_{\text{tot}} = E_\alpha + E_D = (M_\alpha + M_D)c^2 + Q \tag{2.6}$$

5.1.2 Finding p^2c^2

The exact two-body relativistic kinematics means we can solve for p exactly.

We square for both sides of $E_{\text{tot}} = E_\alpha + E_D$

$$E_{\text{tot}}^2 = E_\alpha^2 + E_D^2 + 2E_\alpha E_D$$

We can expand this by using (2.2)

$$E_{\text{tot}}^2 = (p^2c^2 + M_\alpha^2c^4) + (p^2c^2 + M_D^2c^4) + 2\sqrt{(p^2c^2 + M_\alpha^2c^4)(p^2c^2M_D^2c^4)}$$

We can combine the p^2c^2 terms

$$E_{\text{tot}}^2 = 2p^2c^2 + (M_\alpha^2 + M_D^2)c^4 + 2\sqrt{(p^2c^2 + M_\alpha^2c^4)(p^2c^2M_D^2c^4)}$$

We isolate the square root term

$$2\sqrt{(p^2c^2 + M_\alpha^2c^4)(p^2c^2M_D^2c^4)} = E_{\text{tot}}^2 - 2p^2c^2 - (M_\alpha^2 + M_D^2)c^4$$

We square both sides

$$4(p^2c^2 + M_\alpha^2c^4)(p^2c^2M_D^2c^4) = \Big(E_{\text{tot}}^2 - 2p^2c^2 - (M_\alpha^2 + M_D^2)c^4\Big)^2$$

Let $x = p^2c^2$ to clean up the expression. We find that

$$4(x + M_\alpha^2c^4)(xM_D^2c^4) = \Big(E_{\text{tot}}^2 - 2x - (M_\alpha^2 + M_D^2)c^4\Big)^2$$

We expand the LHS

$$4\Big(x^2 + x(M_\alpha^2 + M_D^2)c^4 + M_\alpha^2M_D^2c^8\Big) = \Big(E_{\text{tot}}^2 - 2x - (M_\alpha^2 + M_D^2)c^4\Big)^2$$

Let $y = E_{\text{tot}}^2 - (M_\alpha^2 + M_D^2)c^4$, then we find the RHS to be

$$(y - 2x)^2 = y^2 - 4yx + 4x^2$$

Expanding the LHS, we find

$$4x^2 + 4x(M_\alpha^2 + M_D^2)c^4 + 4M_\alpha^2M_D^2c^8 = y^2 - 4yx + 4x^2$$

Cancelling the $4x^2$, we have

$$4x(M_\alpha^2 + M_D^2)c^4 + 4M_\alpha^2M_D^2c^8 = y^2 - 4yx$$

Moving all x terms to the LHS, we have

$$x\Big(4(M_\alpha^2 + M_D^2)c^4 + 4y\Big) = y^2 - 4M_\alpha^2M_D^2c^8$$

Solving for x, we have

$$x = \frac{y^2 - 4M_\alpha^2M_D^2c^8}{4(M_\alpha^2 + M_D^2)c^4 + 4y}$$

We can simplify the denominator to be

$$\begin{aligned}4(M_\alpha^2 + M_D^2)c^4 + 4y &= 4(M_\alpha^2 + M_D^2)c^4 + 4(E_{\text{tot}}^2 - (M_\alpha^2 + M_D^2)c^4) \\ &= 4E_{\text{tot}}^2\end{aligned}$$

Hence, we have

$$x = \frac{y^2 - 4M_\alpha^2 M_D^2 c^8}{4E_{\text{tot}}^2}$$

Since $y = E_{\text{tot}}^2 - (M_\alpha^2 + M_D^2)c^4$, then the numerator becomes

$$\Big(E_{\text{tot}}^2 - (M_\alpha^2 + M_D^2)c^4\Big)^2 - 4M_\alpha^2 M_D^2 c^8$$

We can factor this as a difference of squares and simplify:

$$\begin{aligned}RHS &= \Big(E_{\text{tot}}^2 - (M_\alpha^2 + M_D^2)c^4 - 2M_\alpha M_D c^4\Big)\Big(E_{\text{tot}}^2 - (M_\alpha^2 + M_D^2)c^4 + 2M_\alpha M_D c^4\Big) \\ &= \Big(E_{\text{tot}}^2 - (M_\alpha + M_D)c^4\Big)\Big(E_{\text{tot}}^2 - (M_\alpha - M_D)c^4\Big)\end{aligned}$$

Hence, we have

$$p^2c^2 = \frac{(E_{\text{tot}}^2 - (M_\alpha + M_D)c^4)(E_{\text{tot}}^2 - (M_\alpha - M_D)c^4)}{4E_{\text{tot}}^2} \tag{2.7}$$

5.1.3 Solving for T_α

We can substitute (2.7) into (2.3), so

$$T_\alpha = \sqrt{\frac{(E_{\text{tot}}^2 - (M_\alpha + M_D)c^4)(E_{\text{tot}}^2 - (M_\alpha - M_D)c^4)}{4E_{\text{tot}}^2} + M_\alpha^2 c^4} - M_\alpha c^2 \tag{2.8}$$

We start work simplifying the expression in the square root. First, we put it to a common denominator

$$\frac{(E_{\text{tot}}^2 - (M_\alpha + M_D)c^4)(E_{\text{tot}}^2 - (M_\alpha - M_D)c^4) + 4E_{\text{tot}}^2 M_\alpha^2 c^4}{4E_{\text{tot}}^2}$$

Let $A = E_{\text{tot}}^2$ and $B = c^4$ to save writing, we find that the numerator is

$$\Big(A - (M_\alpha + M_D)^2 B\Big)\Big(A - (M_\alpha - M_D)^2 B\Big) + 4E_{\text{tot}}^2 M_\alpha^2 BA$$

We expand the product

$$= A^2 - AB\Big((M_\alpha + M_D)^2 + (M_\alpha - M_D)^2\Big) + B^2(M_\alpha + M_D)^2(M_\alpha - M_D)^2 + 4M_\alpha^2 BA$$

We simplify the coefficents

$$(M_\alpha + M_D)^2 + (M_\alpha - M_D)^2 = 2(M_\alpha^2 + M_D^2)$$

So, the AB term becomes

$$\begin{aligned}-AB\Big((M_\alpha + M_D)^2 + (M_\alpha - M_D)^2\Big) + 4ABM_\alpha^2 &= -2AB(M_\alpha^2 + M_D^2) + 4ABM_\alpha^2 \\ &= 2AB\Big(-(M_\alpha^2 + M_D^2) + 2M_\alpha^2\Big) \\ &= 2AB(M_\alpha^2 - M_D^2)\end{aligned}$$

And the B^2 term simplifies to

$$\begin{aligned}(M_\alpha + M_D)^2(M_\alpha - M_D)^2 &= ((M_\alpha + M_D)(M_\alpha - M_D))^2 \\ &= \left(M_\alpha^2 - M_D^2\right)^2\end{aligned}$$

So the entire numerator becomes

$$A^2 + 2AB(M_\alpha^2 - M_D^2) + B^2\left(M_\alpha^2 - M_D^2\right)^2$$

Recognize that this is a perfect square

$$A^2 + 2AB(M_\alpha^2 - M_D^2) + B^2\left(M_\alpha^2 - M_D^2\right)^2 = \left(A + (M_\alpha^2 - M_D^2)B\right)^2$$

That is,

$$\left(E_{\text{tot}}^2 + (M_\alpha^2 - M_D^2)c^4\right)^2$$

So the entire square root term becomes

$$\begin{aligned}\sqrt{\frac{(E_{\text{tot}}^2 - (M_\alpha + M_D)c^4)(E_{\text{tot}}^2 - (M_\alpha - M_D)c^4)}{4E_{\text{tot}}^2} + M_\alpha^2 c^4} &= \sqrt{\frac{(E_{\text{tot}}^2 + (M_\alpha^2 - M_D^2)c^4)^2}{4E_{\text{tot}}^2}} \\ &= \frac{E_{\text{tot}}^2 + (M_\alpha^2 - M_D^2)c^4}{2E_{\text{tot}}}\end{aligned}$$

Therefore, (2.8) reduces to

$$T_\alpha = \frac{E_{\text{tot}}^2 + (M_\alpha^2 - M_D^2)c^4}{2E_{\text{tot}}} - M_\alpha c^2$$

Putting the last term over the denominator, we have

$$T_\alpha = \frac{E_{\text{tot}}^2 + (M_\alpha^2 - M_D^2)c^4 - 2E_{\text{tot}} M_\alpha c^2}{2E_{\text{tot}}} \tag{2.9}$$

From this result, we can then simplify it to a similar form as (2.1).

Since

$$E_{\text{tot}} = (M_\alpha + M_D)c^2 + Q,$$

then

$$T_\alpha = \frac{((M_\alpha + M_D)c^2 + Q)^2 + (M_\alpha^2 - M_D^2)c^4 - 2((M_\alpha + M_D)c^2 + Q)M_\alpha c^2}{2((M_\alpha + M_D)c^2 + Q)}$$

We expand the numerator term by term. For the first term, we have

$$\left((M_\alpha + M_D)c^2 + Q\right)^2 = (M_\alpha + M_D)^2 c^4 + 2(M_\alpha + M_D)c^2 Q + Q^2$$

The second term is already simplified. For the second term, we have

$$\left((M_\alpha + M_D)c^2 + Q\right)M_\alpha c^2 = 2M_\alpha c^4(M_\alpha + M_D) + 2M_\alpha c^2 Q$$

We sum them all together

$$(M_\alpha + M_D)^2 c^4 + 2(M_\alpha + M_D)c^2 Q + Q^2 + (M_\alpha^2 - M_D^2)c^4 - 2M_\alpha c^4(M_\alpha + M_D) - 2M_\alpha c^2 Q$$

We group all the terms by c^4 and c^2. First, the c^4 terms

$$(M_\alpha + M_D)^2 + (M_\alpha^2 - M_D^2) - 2M_\alpha(M_\alpha + M_D)$$

Expand all terms

$$M_\alpha^2 + 2M_\alpha M_D + M_D^2 + M_\alpha^2 - M_D^2 - 2M_\alpha^2 2M_\alpha M_D$$

Notice that this all cancel completely

$$M_\alpha^2 + 2M_\alpha M_D + M_D^2 + M_\alpha^2 - M_D^2 - 2M_\alpha^2 2M_\alpha M_D = 0$$

Combining all the c^2 terms, we have

$$2(M_\alpha + M_D)c^2 Q - 2M_\alpha c^2 Q = 2M_D c^2 Q$$

Now, we have only the Q^2 term. Therefore, the numerator is

$$2M_D c^2 Q + Q^2 = Q(2M_D c^2 + Q)$$

Therefore, (2.9) becomes

$$T_\alpha = \frac{Q(2M_D c^2 + Q)}{2((M_\alpha + M_D)c^2 + Q)}$$

This can then be factored to become

$$\boxed{T_\alpha = Q\frac{(M_D c^2 + \frac{Q}{2})}{((M_\alpha + M_D)c^2 + Q)}} \tag{2.10}$$

5.1.4 Does it reduce to the nonrelativistic limit?

As a sanity check, we can check if (2.10) reduces to (2.1) at the nonrelativistic limit. For the nonrelativistic limit, we assume that

$$Q \ll M_\alpha c^2 \qquad\qquad Q \ll M_D c^2$$

So the denominator is approximately

$$(M_\alpha + M_D)c^2 + Q \approx (M_\alpha + M_D)c^2$$

And the numerator is approximately

$$M_D c^2 + Q \approx M_D c^2$$

Therefore, we see that it does in fact reduce to (2.1)!

$$T_\alpha \approx Q\frac{M_D c^2}{(M_\alpha + M_D)c^2} = Q\frac{M_D}{M_\alpha + M_D}$$

5.1.5 Error made by neglecting relativistic corrections

The absolute error for the relativistic and nonrelativistic energies are

$$\Delta T_\alpha = T_\alpha^R - T_\alpha^N$$

Where T_α^R denotes the relativistic kinetic energy and T_α^N denotes the nonrelativistic (Newtonian) kinetic energy. We have

$$\Delta T = Q\frac{(M_D c^2 + \frac{Q}{2})}{((M_\alpha + M_D)c^2 + Q)} - Q\frac{M_D}{M_\alpha + M_D}$$

We simplify. We factor out the Q on both

$$\Delta T = Q\left(\frac{(M_D c^2 + Q)}{(2(M_\alpha + M_D)c^2 + Q)} - \frac{M_D}{M_\alpha + M_D}\right)$$

We place it all under a common denominator $2(M_\alpha + M_D)((M_\alpha + M_D)c^2 + Q)$.

$$\Delta T = Q\frac{(M_\alpha + M_D)(2M_D c^2 + Q) - 2M_D((M_\alpha + M_D)c^2 + Q)}{2(M_\alpha + M_D)((M_\alpha + M_D)c^2 + Q)}$$

We expand the first term of the numerator, we have

$$(M_\alpha + M_D)2M_D c^2 + (M_\alpha + M_D)Q - 2M_D(M_\alpha + M_D)c^2 - 2M_D Q$$

The $2M_D(M_\alpha + M_D)c^2$ terms cancel. We are left with

$$(M_\alpha + M_D)Q - 2M_D Q = (M_\alpha - M_D)Q$$

Therefore, we have the absolute error

$$\boxed{\Delta T = \frac{Q^2(M_\alpha - M_D)}{2(M_\alpha + M_D)((M_\alpha + M_D)c^2 + Q)}}$$

For the relative error, we have

$$\begin{aligned}\frac{\Delta T}{T_\alpha^N} &= \frac{\frac{Q^2(M_\alpha - M_D)}{2(M_\alpha + M_D)((M_\alpha + M_D)c^2 + Q)}}{Q\frac{M_D}{M_\alpha + M_D}} \\ &= \frac{Q(M_\alpha - M_D)}{2((M_\alpha + M_D)c^2 + Q)}\end{aligned}$$

For $Q \ll (M_\alpha + M_D)c^2$, this can be simplified as

$$\frac{\Delta T}{T_\alpha^N} \approx \frac{Q(M_\alpha - M_D)}{2(M_\alpha + M_D)c^2}$$

An alternative form of this (since $M_\alpha - M_D$ is negative) is

$$\frac{\Delta T}{T_\alpha^N} \approx \frac{Q}{2c^2}\left(\frac{1}{M_D} - \frac{1}{M_\alpha + M_D}\right)$$

or

$$\boxed{\frac{\Delta T}{T_\alpha^N} \approx \frac{Q}{2M_D c^2}\left(1 - \frac{M_D}{M_\alpha + M_D}\right)}$$

References

[1] M. Flores. Lecture Slides *Physics 180: Nuclear and Particle Physics*. Digital lecture slides, 2026. Available from course website, UVLé.

Appendices

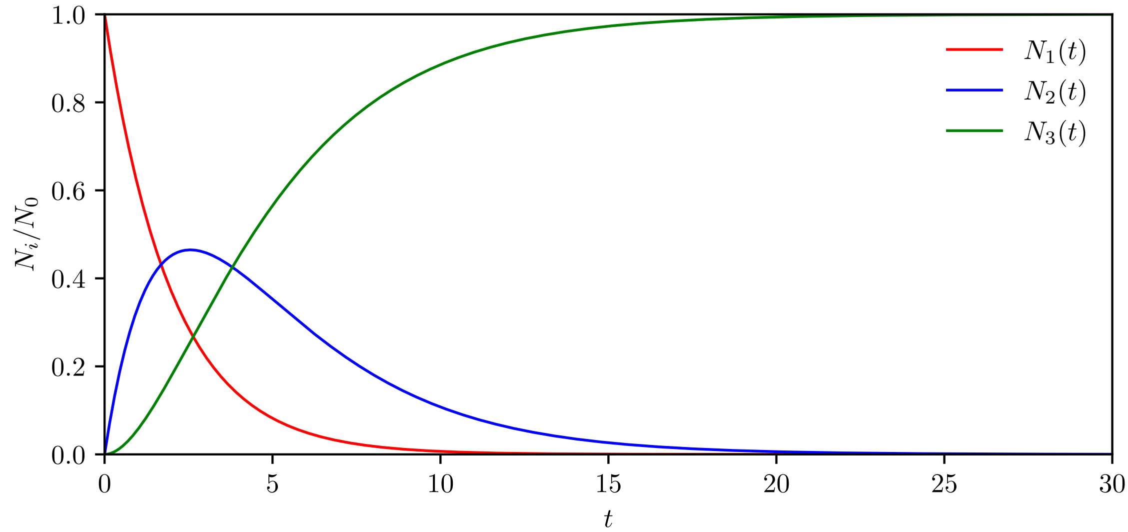

Figure 1: Evolution of the populations in the decay chain $1 \to 2 \to 3$, with initial population N_0. The decay constants are $\lambda_1 = 0.5$ and $\lambda_2 = 0.3$. Type 1 decays into Type 2, which subsequently decays into the stable Type 3 product.

Code

No code was used for this problem set